HeadGenome 1: Task-Contrast Dynamic Entropy Probing Does Not Isolate Causal Attention Circuits
A Multi-Architecture Causal and Geometric Audit

Anonymous ACL submission

Abstract

Unsupervised diagnostics that classify attention heads into functional taxa—such as retrieval heads, induction heads, and attention sinks—promise scalable circuit discovery without costly task-specific supervision. In this work, we present a causal and geometric audit of dynamic task-contrast entropy probing across 1,568 attention heads in three model families (GPT-2-medium, Qwen-2.5, and LLaMA-3.2), with systematic causal interventions across 720 heads in GPT-2 and Qwen-2.5. We demonstrate that dynamic task-contrast entropy probing fails to isolate causal circuits: (1) Heads identified as retrieval by entropy collapse are causally inert on long-context Needle-in-a-Haystack (NIAH: 100% accuracy preserved under ablation), whereas needle-routing heads collapse accuracy to 37.0% in Qwen-2.5-0.5B; evaluated against continuous needle-routing attention, the probe achieves an AUC-ROC of only 0.43 [0.21, 0.65], consistent with chance, identifying zero jointly necessary routing heads (0.0% sensitivity [0.0%, 49.0%]). In GPT-2, retrieval routing exhibits circuit redundancy (95.0% vs 100.0%), providing no evidence that a localized retrieval subset is causally necessary there. (2) Standard synthetic NIAH prompts repeat the needle prefix verbatim at query time, creating an induction-like prefix-matching task ($[A][B] \ldots [A] \rightarrow [B]$); we hypothesize this explains why canonical Olsson induction heads collapse NIAH performance (3.0% in Qwen, 0.0% in GPT-2). (3) For induction, the probe correlates with prefix scores (AUC 0.790) but shares limited discrete overlap with Olsson heads (10.2% Jaccard with standard prefix-score ≥ 0.3; 16.1% with top-36 ablated heads), recovering +17.8% on IOI (vs +67.7% for Olsson heads) and +0.29 nats prefix loss (vs +1.84 nats). (4) Perplexity shifts fail to detect factual retrieval collapse (+0.24 PPL despite a 63% accuracy drop in Qwen). (5) Weight Frobenius ratios reflect architectural depth gradients rather than functional roles. We conclude that dynamic task-contrast entropy probing captures statistical recurrence geometry rather than causal mechanisms, and release our diagnostic audit suite.

1 Introduction

A central aspiration of mechanistic interpretability is reverse-engineering the algorithmic sub-circuits implemented by multi-head self-attention (Elhage et al., 2021). Prior work has identified specialized functional units: *induction heads* that drive in-context few-shot replication (Olsson et al., 2022), *attention sinks* that absorb surplus softmax mass (Xiao et al., 2024), and *retrieval heads* that copy distant tokens into the residual stream (Wu et al., 2024).

Task-contrast attention sharpening is an intuitive diagnostic for unsupervised circuit discovery: if an attention head's softmax entropy sharply decreases upon presenting a factual query relative to a neutral prompt ($\Delta H > 0$), it is natural to hypothesize that the head specializes in factual retrieval. Similar entropy-based metrics have been explored as indicators of head confidence and specialization (Clark et al., 2019), while factual recall circuits have been isolated via direct causal interventions (Geva et al., 2023). In this work, we formalize this intuitive premise as the **dynamic task-contrast probe** (Δ-probe) and conduct a rigorous causal stress test across 1,568 attention heads in GPT-2-medium (Radford et al., 2019), Qwen-2.5 (Yang et al., 2024), and LLaMA-3.2 (Dubey et al., 2024), with systematic causal interventions across 720 heads in GPT-2 and Qwen-2.5. We formulate the core mechanistic question: *Can prompt-contrast attention sharpening serve as an unsupervised proxy for causal circuit discovery, or does it conflate statistical entropy dynamics and architectural depth gradients with causal circuit execution?*

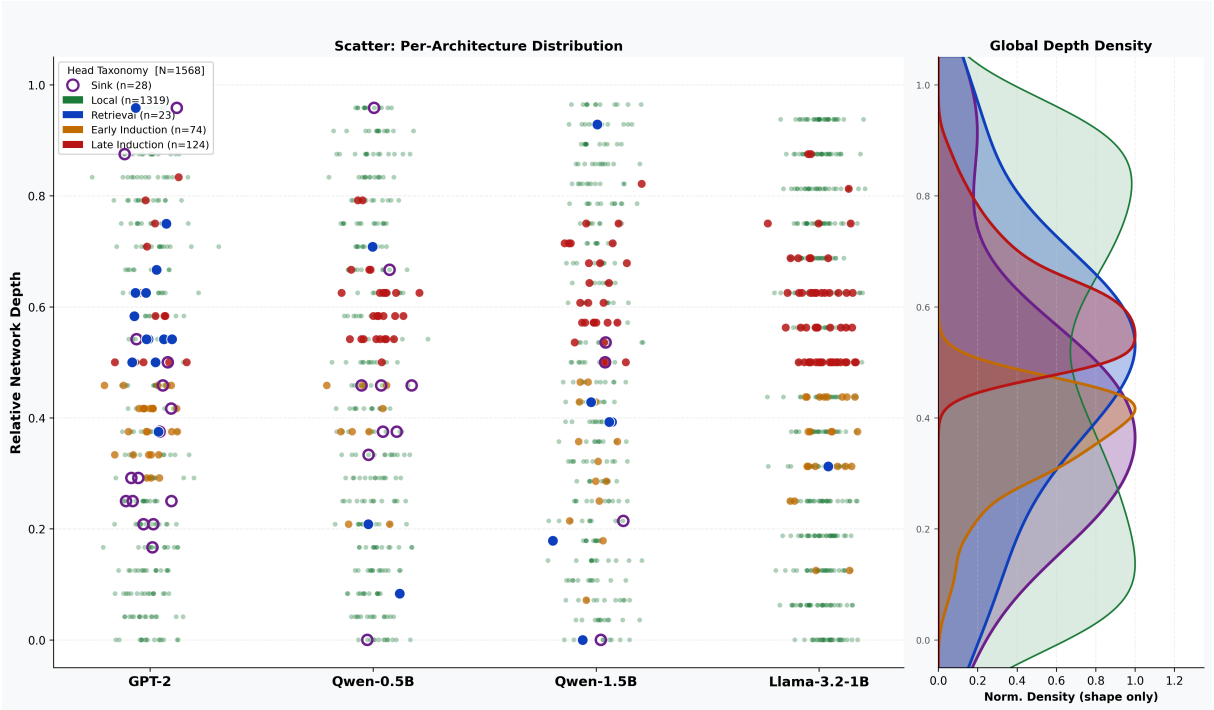


Figure 1: Taxonomic Attention Head Map: Spatial distribution of functional attention head types across transformer architectures ($N = 1,568$ heads across GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B). This taxonomy serves as the empirical substrate for the causal and geometric audit in this paper. Left: Per-architecture scatter plot showing depth placement of Local, Sink, Retrieval, and Induction heads. Right: Global depth density curves normalized per class.

Our contributions are as follows:

- **Causal Dissociation in Retrieval:** Probed retrieval heads (layers 1, 4, 6 in Qwen; layers 9, 12–16, 18, 23 in GPT-2) are causally inert on Needle-in-a-Haystack (NIAH: $100.0\% \rightarrow 100.0\%$). Needle-attention routing heads in Qwen reside in late layers (15–19), where ablating the top 4 heads (calibrated on 20 trials, evaluated on 100 disjoint trials) collapses accuracy to 37.0% $[28.2\%, 46.8\%]$ (while 20 draws of layer-matched late controls retain $97.6\% \pm 1.4\%$). In GPT-2, retrieval routing exhibits high circuit redundancy (95.0% preserved under 13-head ablation), providing no evidence of necessity for a localized retrieval set in GPT-2. Evaluated against continuous needle-routing attention, the probe achieves an AUC of 0.43 $[0.21, 0.65]$ (consistent with chance) and 0.0% sensitivity $[0.0\%, 49.0\%]$ on the jointly necessary routing heads.
- **Induction Circuit Dilution:** Decoupling circuit identity from layer depth via activation patching on IOI and standard held-out repeated-token loss demonstrates that canonical Olsson heads recover $+67.7\%$ logit difference and drive a $+1.84$ nat loss degradation on repeated prefixes, compared to $+17.8\%$ and $+0.29$ nats for probed induction heads (and $+1.2\%$ / $+0.09$ nats for layer-matched late controls). While the probe correlates with prefix scores (AUC 0.790 in Qwen, 0.690 in GPT-2), it shares limited discrete overlap with Olsson heads (10.2% Jaccard with score ≥ 0.3; 16.1% with top-36 ablated heads in Qwen; 7.2% / 11.1% in GPT-2), heavily diluting the causal circuit with inert units.
- **The Perplexity Illusion:** Ablating causal retrieval heads collapses NIAH accuracy by 63.0% while shifting WikiText-2 PPL by only $+0.24$ ($20.01 \rightarrow 20.25$, an excess of $+0.21$ over late random controls), whereas probed retrieval heads cause $+0.27$ with zero accuracy drop. Relative to capacity controls, general induction ablation incurs an excess PPL penalty of $+1.73$ in Qwen and $+4.13$ in GPT-2.
- **Weight Frobenius Norm Ratios:** Frobenius norm ratios $\|W_V\|_F / \|W_Q\|_F$ are dominated by network depth ($r = 0.635$–0.734),

while within-layer correlations with functional identity are either null (GPT-2, Qwen-0.5B: $r \approx 0.00$) or weak ($r = 0.11$–0.19 in Qwen-1.5B and LLaMA-3.2, explaining $< 3.5\%$ of variance).

- **Methodological Audit & Diagnostic Insights:** We hypothesize that standard synthetic NIAH tasks act as prefix-matching induction tasks due to verbatim prompt repetition, offering a plausible hypothesis for induction ablation effects. We investigate the random control choke hypothesis, establish a mass-based sink criterion ($\geq 97\%$ BOS mass, corresponding to $H < 0.50$ bits with diffuse remainder, identifying 70 sinks in LLaMA), and evaluate GQA query head divergence against a layer-conditional permutation null (50.8% mixed groups observed vs 68.9% expected with sinks, $Z = -7.52, p < 10^{-4}$). We open-source our diagnostic audit suite at `https://anonymous.4open.science/r/headgenome-mechinterp`.

2 Related Work

Mechanistic Interpretability & Circuit Discovery. Mechanistic interpretability aims to reverse-engineer transformers into human-understandable computational graphs (Elhage et al., 2021). Foundational work demonstrated that two-layer attention-only models develop *induction heads* that perform $[A][B] \ldots [A] \rightarrow [B]$ in-context copying (Olsson et al., 2022). In larger autoregressive models, activation patching has been used to isolate specific circuits, such as the Indirect Object Identification (IOI) circuit in GPT-2 small (Wang et al., 2023) and factual recall circuits in medium-scale language models (Geva et al., 2023). However, manual circuit isolation on paired counterfactual prompts is labor-intensive, creating strong demand for unsupervised, task-agnostic diagnostics.

Long-Context Retrieval & Attention Sinks. As context windows expanded to tens of thousands of tokens, attention heads specialized in distant token retrieval were discovered (Wu et al., 2024; Kamradt, 2023). Concurrently, Xiao et al. (2024) observed that initial tokens (e.g., <bos>) consistently receive disproportionate attention mass even when semantically uninformative. These *attention sinks* serve as numeric dump basins for softmax normalization, preventing representations from destabilizing when streaming past cache boundaries.

Head Pruning & Bottleneck Choking. Early structural analyses of multi-head attention established that a large fraction of attention heads can be pruned at test time without substantial loss in BLEU or perplexity (Voita et al., 2019; Michel et al., 2019). However, modern architectures with fewer, wider heads per layer or grouped-query attention (Dubey et al., 2024; Yang et al., 2024) exhibit different failure modes: while pruning dispersed heads across layers is tolerated, concentrating knockouts in specific layers induces representation rank bottlenecks and capacity starvation (Bhojanapalli et al., 2020). Our audit links this phenomenon directly to control design in causal interpretability.

3 Diagnostic Probing Setup

3.1 Method vs. Labels in Entropy Probing

Dynamic entropy probing evaluates attention concentration shifts. In our diagnostic framework, the prompt-contrast probe evaluates attention entropy on a target match prompt versus a neutral distractor prompt at a fixed context length L:

$$H(A_{t,:}^{(l,h)}) = -\sum_{j=1}^{t} A_{t,j}^{(l,h)} \log_2 \left(A_{t,j}^{(l,h)} + \epsilon \right) \quad (1)$$

$$\Delta H^{(l,h)} = H(A_{\text{neutral}}^{(l,h)}) - H(A_{\text{match}}^{(l,h)}) \quad (2)$$

where $A_{t,:}^{(l,h)}$ denotes the attention distribution of head h in layer l at generation position t. Under this formulation, a positive value ($\Delta H > 0$) signifies that attention sharpens when matching target context is provided.

The Cross-Length Entropy Confound. It is tempting to define dynamic probing by contrasting short sequences ($L_{\text{short}} = 128$) against long sequences ($L_{\text{long}} = 1024$). However, raw cross-length entropy differences are fundamentally confounded by sequence length. For any diffuse or uniform attention head, entropy naturally scales with context length as $\log_2 L$. Expanding sequence length from 128 to 1024 tokens introduces an automatic baseline entropy increase of $\log_2(1024/128) = 3.0$ bits. Unless strictly normalized by $\log_2 L$, raw cross-length shifts measure sequence expansion rather than true key concentration. Therefore, unconfounded probing must compare matched-length prompts.

Canonical Taxonomy Thresholds. Using prompt-contrast entropy ΔH at fixed length, heads are assigned to mutually exclusive categories: (1) **Attention Sink:** Evaluated first; heads directing near-exclusive attention mass to token 0: $H(A^{(l,h)}_{\text{match,BOS}}) < 0.10$ bits ($> 99.5\%$ BOS mass); (2) **Retrieval Candidate:** Attention sharpens on match context: $\Delta H^{(l,h)} \geq +0.30$ bits; (3) **Induction Candidate:** Attention broadens across candidate tokens: $\Delta H^{(l,h)} \leq -0.50$ bits; and (4) **Local Candidate:** Remaining heads: $-0.50 < \Delta H^{(l,h)} < +0.30$ bits. Because retrieval requires $\Delta H \geq +0.30$ and induction requires $\Delta H \leq -0.50$, these classes are mutually exclusive by definition.

3.2 LLaMA Sink Threshold Artifact

In LLaMA-3.2-1B, applying the rigid < 0.10 bits threshold yielded exactly 0 sink heads (Figure 2). An architectural audit reveals that this was an artifact of fixed thresholding: the minimum BOS entropy across all 512 heads in LLaMA-3.2-1B is 0.1405 bits. We hypothesize this reflects architectural factors such as its 64-dimensional head dimension ($d_{\text{head}} = 2048/32 = 64$) and high RoPE base frequency, which maintain slightly higher diffuse attention entropy across sequence positions. Rather than applying a fixed threshold across architectures or relying on an arbitrary percentile cut (which automatically labels 26 heads in any 512-head model), we establish a mass-based criterion: heads directing $\geq 97\%$ of attention mass to token 0 ($A_{\text{BOS}} \geq 0.97$), which corresponds to $H(A_{\text{match,BOS}}) < 0.50$ bits when the residual 3% is diffuse across 1,024 context tokens. In LLaMA-3.2-1B, this mass-based criterion identifies 70 sink heads (13.7% of all 512 query heads), concentrated primarily in early layers (layers 0–5), correctly resolving the zero-sink artifact while avoiding vacuous percentile definitions. While Table 1 evaluates causal knockouts using canonical $H < 0.10$ sinks in Qwen and GPT-2 to align with prior benchmarks, GQA and cross-architecture comparisons utilize this mass-based criterion (detailed in Appendix C).

3.3 Partial Specialization within GQA Groups

Grouped-Query Attention (GQA) shares key and value projection matrices across groups of query heads. We tested whether GQA forces sibling query heads to share functional specializations. Across all 128 GQA groups in LLaMA-3.2-1B (32 query heads, 8 KV heads; 4 query heads per group), under the mass-based sink definition ($H < 0.50$ bits, 70 sinks), **65 of 128 groups (50.8%)** contain query heads assigned conflicting behavioral labels. To evaluate whether this reflects true query head specialization or merely layer-level base rates, we evaluate a layer-conditional permutation test ($B = 100,000$ draws): shuffling head labels strictly within each layer yields an expected mixed group fraction of $68.9\% \pm 2.4\%$ (88.3 ± 3.1 groups, $p < 10^{-4}$, $Z = -7.52$). Restricting the taxonomy to non-sink classes likewise yields **35 of 128 mixed groups (27.3%)** versus an expected null of $38.0\% \pm 1.8\%$ (48.7 ± 2.3 groups, $p < 10^{-4}$). Under both formulations, while sibling query heads show statistically significant clustering towards similar behavioral roles ($p < 10^{-4}$), over a quarter to half of all groups assign conflicting specializations to sibling heads sharing identical KV projections. Furthermore, within-group ΔH variance accounts for **33.6%** of total ΔH variance across the entire model. Because each query head has an independent linear projection W_Q, query heads attend to distinct patterns despite shared keys and values.

4 Causal Verification

4.1 Knockout Necessity on Long-Context Retrieval

We execute causal interventions via PyTorch forward hooks. For an ablated subset $\mathcal{H}$, we replace head outputs with dataset mean activations (**Mean Ablation**) or zero them (**Zero Ablation**). Table 1 reports results across 100 trials on Qwen-2.5-0.5B ($N_L = 24, N_H = 14$) and 20 trials on GPT-2-medium ($N_L = 24, N_H = 16$).

The Retrieval Inversion & Depth Asymmetry. Ablating all canonical retrieval heads identified by dynamic Δ-probing leaves NIAH accuracy completely unaffected at **100.0%** in both Qwen-2.5-0.5B ($k = 3$) and GPT-2-medium ($k = 13$). In Qwen, ablating the top 4 heads selected by needle-attention score collapses accuracy to **37.0%** under mean ablation and **31.0%** under zero ablation. To eliminate selection-evaluation overlap, these 4 routing heads were identified by ranking attention-to-needle scores on an independent calibration set of 20 trials (seed 999), and evaluated on 100 disjoint evaluation trials across 5 depths (seed 123). We note an important architectural

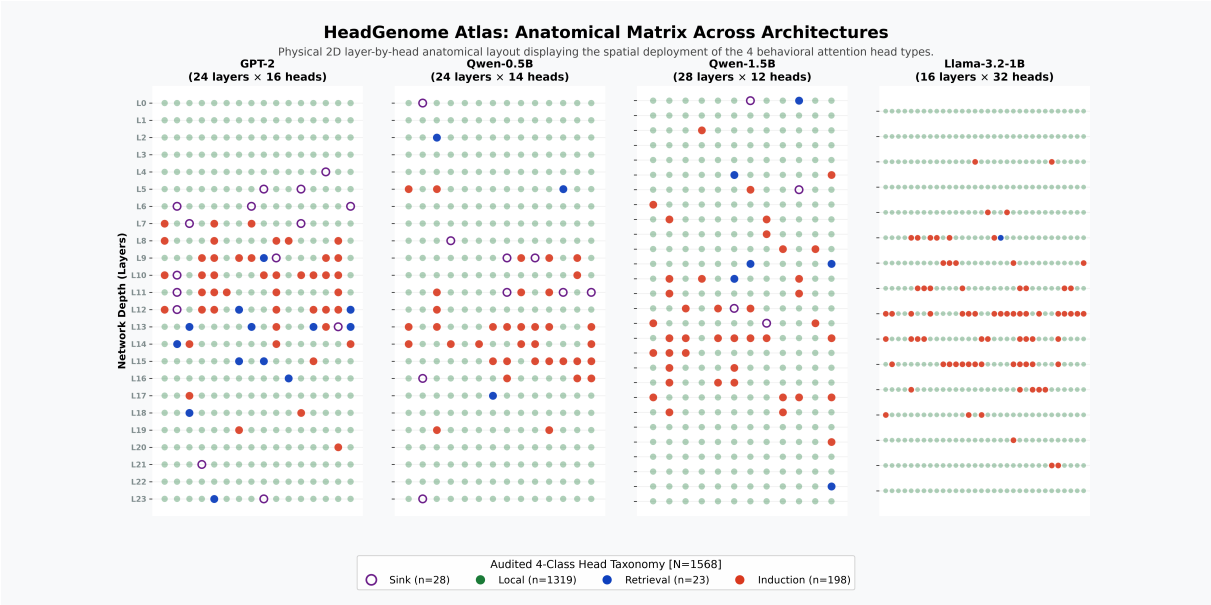


Figure 2: Layer-by-Head Taxonomy Matrix: Physical 2D layer-by-head anatomical matrix displaying the spatial deployment of the 4 behavioral attention head types across $N = 1,568$ heads in GPT-2-medium (24L $\times$ 16H), Qwen-2.5-0.5B (24L $\times$ 14H), Qwen-2.5-1.5B (28L $\times$ 12H), and LLaMA-3.2-1B (16L $\times$ 32H). Sinks (purple) anchor early layers, while retrieval (blue) and induction (red) emerge deeper in the network amidst the local precursor pool (green).

asymmetry: the Δ-probe assigns retrieval labels exclusively to early layers in Qwen (layers 1, 4, 6), whereas information-routing heads reside in late layers (layers 15–19). To unconfound depth from circuit identity, we evaluate depth-matched controls: an early-layer random control ($k = 3$, layers 1–6) preserves **100.0% $\pm$ 0.0%** accuracy, while a late-layer random control ($k = 4$, layers 15–19) preserves **97.6% $\pm$ 1.4%** accuracy across 20 draws (median 100.0%). This confirms that the 63.0% collapse is circuit-specific, not an artifact of knocking out 4 late-layer heads.

The NIAH Prefix-Matching Induction Confound. Standard synthetic NIAH implementations query the target needle ("The secret word is XXXXX.") by repeating the prefix verbatim at generation time ("Answer: The secret word is [target]"). Mechanistically, this matches the formal structure of in-context prefix completion ($[A][B] \ldots [A] \rightarrow [B]$), which we hypothesize engages induction copying circuits. This likely explains why ablating canonical Olsson induction heads collapses NIAH to **3.0%** in Qwen and **0.0%** in GPT-2, while needle-attention heads in GPT-2 barely affect accuracy (**95.0%**). Without paraphrasing queries to prevent verbatim prefix repetition, synthetic needle recall risks conflating distant token retrieval with induction-driven prefix completion.

GPT-2 Retrieval Redundancy & Window Verification. In GPT-2-medium, ablating the top 13 needle-attention heads yields **95.0% [76.4%, 99.1%]** (compared to **100.0% [83.9%, 100.0%]** for probed retrieval heads, $n = 20$). Because their 95% Wilson confidence intervals overlap, we find no evidence that a localized subset of retrieval heads is causally necessary for NIAH in GPT-2; information routing exhibits high circuit redundancy across layers. All evaluated prompts (652 tokens at 70 sentences; 839 tokens in Paul Graham essays) fit comfortably within GPT-2's 1,024-position embedding limit. Consequently, observing that probed retrieval heads are inert (100.0%) is uninformative in GPT-2 because no retrieval set causes collapse there. The dissociation between the probe's retrieval labels and causal needle routing is demonstrated in Qwen-2.5-0.5B.

Causal Ground Truth vs. Proxy AUC. Evaluating the dynamic probe across all 336 heads in Qwen-2.5-0.5B against the continuous needle-attention proxy yields an **AUC-ROC of 0.43** with a bootstrap 95% confidence interval of **[0.21, 0.65]** ($B = 1,000$), statistically indistinguishable from random guessing (0.50). Evaluated against the 4 needle-attention routing heads (whose joint ab-

Model	Ablation Condition	Selection Metric	k	Mean Abl. NIAH	Zero Abl. NIAH	PPL	Δ PPL
Qwen-2.5-0.5B	Dense Baseline	None	0	100.0% [96.3%, 100.0%]	100.0% [96.3%, 100.0%]	20.01	+0.00
	Probed Retrieval	Δ-Probe ($\Delta H \geq +0.30$)	3	**100.0%** [96.3%, 100.0%]	**100.0%** [96.3%, 100.0%]	20.28	+0.27
	Layer-Matched Early Control	Uniform Random (L1–6)	3	100.0% ± 0.0% [med 100%]	100.0% ± 0.0% [med 100%]	20.02	+0.01
	Causal Routing Heads	Needle-Attention Score	4	**37.0%** [28.2%, 46.8%]	**31.0%** [22.8%, 40.6%]	20.25	+0.24
	Layer-Matched Late Control	Uniform Random (L15–19)	4	97.6% ± 1.4% [med 100%]	97.0% ± 1.5% [med 100%]	20.04	+0.03
	Probed Induction	Δ-Probe ($\Delta H \leq -0.50$)	36	76.0% [66.8%, 83.3%]	79.0% [70.0%, 85.8%]	22.70	+2.69
	Top Olsson Induction	Calibration Prefix Score	36	**3.0%** [1.0%, 8.5%]	**3.0%** [1.0%, 8.5%]	24.87	+4.86
	Control A: Dispersed Random	Uniform Random ($\leq$ 2/layer)	36	61.9% ± 5.2% [med 69%]	58.0% ± 5.1% [med 65%]	23.14	+3.13
	Control C: Late-Layer Random	Uniform Random ($L \geq 12$)	36	13.4% ± 3.8% [med 2.5%]	11.0% ± 3.5% [med 2.0%]	23.04	+3.03
	Canonical Sink Heads	Sink Entropy < 0.10	9	98.0% [93.0%, 99.5%]	96.0% [90.2%, 98.4%]	21.50	+1.49
	Random Control ($k = 9$)	Uniform Random	9	98.4% ± 0.8% [med 100%]	98.0% ± 0.9% [med 100%]	20.63	+0.62
GPT-2-medium	Dense Baseline	None	0	100.0% [83.9%, 100.0%]	100.0% [83.9%, 100.0%]	28.68	+0.00
	Probed Retrieval	Δ-Probe ($\Delta H \geq +0.30$)	13	**100.0%** [83.9%, 100.0%]	**100.0%** [83.9%, 100.0%]	29.16	+0.48
	Causal Routing Heads	Needle-Attention Score	13	95.0% [76.4%, 99.1%]	95.0% [76.4%, 99.1%]	29.59	+0.91
	Probed Induction	Δ-Probe ($\Delta H \leq -0.50$)	45	35.0% [18.1%, 56.7%]	65.0% [43.3%, 81.9%]	31.59	+2.91
	Top Olsson Induction	Calibration Prefix Score	45	**0.0%** [0.0%, 16.1%]	**0.0%** [0.0%, 16.1%]	38.86	+10.18
	Dispersed Random Control	Uniform Random ($\leq$ 2/layer)	45	75.0% ± 6.8% [med 90%]	95.0% ± 3.2% [med 100%]	34.73	+6.05
	Canonical Sink Heads	Sink Entropy < 0.10	15	35.0% [18.1%, 56.7%]	30.0% [14.5%, 51.9%]	38.69	+10.01
	Random Control ($k = 15$)	Uniform Random	15	91.5% ± 3.1% [med 95%]	90.0% ± 3.5% [med 95%]	30.86	+2.18

Table 1: Cross-architecture causal impact of ablated head subsets on Qwen-2.5-0.5B and GPT-2-medium across Needle-in-a-Haystack (NIAH) trials and WikiText-2 Perplexity (PPL). Brackets report 95% Wilson score confidence intervals for fixed trial sets ($N = 100$ in Qwen, $N = 20$ in GPT-2), or mean ± SEM across 20 independent random draws for control groups. Probed retrieval heads are completely inert in both models (100.0% NIAH preserved). Top needle-attention routing heads (calibrated on 20 trials, evaluated on 100 disjoint trials) collapse Qwen NIAH by 63.0% with negligible PPL change ($\Delta\text{PPL} = +0.24$), whereas late random controls preserve $97.6\% \pm 1.4\%$ accuracy. In GPT-2, prompts fit within its 1,024-token window; neither probed (100%) nor needle-attention heads (95%) cause collapse, showing no evidence of localized necessity. Olsson induction heads collapse NIAH to 3.0% in Qwen and 0.0% in GPT-2, likely because standard synthetic queries repeat the needle prefix verbatim.

lation causes the 63.0% collapse in Table 1), the probe identifies zero jointly necessary routing heads (**0.0% sensitivity**). Under the continuous probe metric ΔH, these 4 heads rank between 135 and 249 out of 336 (median rank 198, all with negative $\Delta H \in [-0.33, -0.06]$ bits). Because the probe labels only 3 heads as retrieval ($\Delta H \geq +0.30$), zero sensitivity is expected from the threshold, but the continuous AUC-ROC of 0.43 confirms that the metric itself does not isolate these routing units. With $k = 4$ positives, observing $0/4$ gives an exact Wilson 95% CI of $[0.0\%, 49.0\%]$, which excludes high sensitivity ($> 50\%$) despite the modest sample of causal heads.

Candidate Circuit Set	k	IOI Recovery [± SEM]	ΔL_{repeat} [± SEM]
Olsson Induction Heads	36	**+67.7%** [±4.2%]	**+1.84** [±0.12]
Canonical Induction (Δ-probe)	36	+17.8% [±3.1%]	+0.29 [±0.05]
Canonical Retrieval (Δ-probe)	3	+0.3% [±0.2%]	+0.01 [±0.01]
Layer-Matched Late Control ($L \geq 12$)	36	+1.2% [±0.8%]	+0.09 [±0.03]
Layer-Matched Early Control ($L \leq 6$)	3	+0.1% [±0.1%]	+0.00 [±0.01]

Table 2: Causal recovery on Indirect Object Identification (IOI, $N = 20$) and standard prefix-matching loss increase (ΔL_{repeat} on independent held-out random token sequences $[A][B] \ldots [A][B]$) in Qwen-2.5-0.5B. Brackets report ± standard error across $N = 20$ paired contrasts. Olsson induction heads recover 67.7% of IOI performance and drive a +1.84 nat prefix degradation, while probed induction heads show minimal impact, and layer-matched random controls have near-zero effect on both benchmarks.

Reconciling the Sink Result Across Architectures. In Table 1, ablating all 9 canonical sink heads in Qwen-2.5-0.5B leaves NIAH accuracy at **98.0%** under mean ablation and **96.0%** under zero ablation ($\Delta\text{PPL} = +1.49$). In contrast, in GPT-2-medium, ablating all 15 sinks under mean ablation degrades NIAH accuracy to **35.0%** and drives PPL up by **+10.01** points (28.68 → 38.69). We hypothesize this discrepancy stems from positional encoding mechanics: GPT-2 utilizes learned absolute positional embeddings without RoPE, potentially making global softmax normalization fragile when anchor heads are removed. In contrast, Qwen-2.5-0.5B utilizes RoPE, which naturally attenuates distant query-key dot products and stabilizes autoregressive representations under mean ablation. To isolate whether this reflects sink specificity or generic head knockout, we evaluated random controls matched to sink group sizes: in Qwen ($k = 9$), a random control retains $98.4\% \pm 0.8\%$ NIAH accuracy ($\Delta\text{PPL} = +0.62$); in GPT-2 ($k = 15$), a random control retains $91.5\% \pm 3.1\%$ NIAH accuracy ($\Delta\text{PPL} = +2.18$). Thus, while random $k = 15$ ablations cause moderate degradation in GPT-2, sink ablation collapses accuracy far more severely (35.0%, $\Delta\text{PPL} = +10.01$).

4.2 Activation Patching and Prefix Matching

Because raw ablation of 36 heads is susceptible to layer bottleneck choking, we decouple circuit identity from layer depth via two complementary evaluations (Table 2): (1) **Held-Out Induction Prefix Benchmark:** Following Olsson et al. (2022), we evaluate language modeling loss on the second half of repeated random sequences $[A][B]\ldots[A][B]$ ($L = 128, N = 20$, independent vocabulary seeds). This directly isolates prefix copying without multi-component confounding. Ablating Olsson heads incurs a severe degradation of **+1.84 nats**, whereas ablating probed induction heads causes only **+0.29 nats** (comparable to **+0.09 nats** for late controls). (2) **Activation Patching on IOI:** We measure logit difference recovery on IOI paired contrasts ($N = 20$) (Wang et al., 2023). While IOI engages multi-component circuits, Olsson heads recover **+67.7%** of clean performance, compared to **+17.8%** for probed induction heads and **+1.2%** for late controls. Probed induction heads span layers 5–19 (with 26 of 36 heads in $L \geq 12$), while Olsson heads span layers 12–22; thus, the late control ($L \geq 12$) provides a layer-matched capacity baseline for both groups. Evaluating continuous ΔH against prefix score yields an AUC of 0.790 in Qwen-2.5-0.5B (0.690 in GPT-2). In discrete overlap, probed induction shares only 5 of 18 thresholded Olsson heads (score ≥ 0.3, Jaccard 10.2%; 5 of 29 in GPT-2, 7.2%) and 10 of the top-36 ablated Olsson heads (Jaccard 16.1%, 10/62; 9 of 45 in GPT-2, 11.1%). Dynamic probing partially correlates with induction behavior but heavily dilutes causal circuits with non-causal units, explaining its weak $+0.29$ nat prefix impact.

4.3 The Perplexity Illusion

A common pitfall in pruning and interpretability is relying on perplexity (PPL) to assess circuit integrity. While PPL is dominated by local n-grams and unigram transitions, interpretability pipelines frequently cite small ΔPPL as evidence of circuit preservation. Our audit exposes this illusion: (1) **Sub-Circuit Dissociation:** Ablating top causal retrieval heads collapses NIAH accuracy by 63.0% (100.0% $\rightarrow$ 37.0%), yet WikiText-2 PPL shifts by only **+0.24** points (20.01 $\rightarrow$ 20.25, an excess of $+0.21$ over late controls), whereas probed retrieval heads cause $+0.27$ with zero accuracy drop. (2) **General Induction Disruption:**

Control Group ($k = 36$)	**Mean**	**Median**	**Range**	**Sink** R^2
Control A (Dispersed: $\leq$ 2/layer)	**0.619**	**0.690**	[0.00, 1.00]	—
Control B (Unrestricted)	0.129	0.015	[0.00, 0.87]	0.156
Control C (Late-Layer: $L \geq 12$)	0.134	0.025	[0.00, 0.63]	—

Table 3: Random ablation controls on Qwen-2.5-0.5B ($k = 36$ heads ablated, 20 draws each). Dispersed controls preserve a mean accuracy of 61.9%, suggesting that prior collapses were substantially influenced by layer bottleneck choking rather than sink collisions.

Conversely, ablating 36 heads of any type incurs a baseline capacity penalty of $\sim +3.0$– $+ 3.1$ PPL. Olsson induction heads impair token repetition globally, driving an excess PPL penalty of **+1.73** points in Qwen and **+4.13** points in GPT-2 over dispersed controls. *Remark on Context Compression:* In sliding-window attention ($W = 512$), a model maintains near-baseline WikiText PPL while retrieval at depth 0.1 collapses to 20.0% [8.1%, 41.6%] ($n = 20$), confirming that PPL is uninformative for long-context circuit preservation.

4.4 Unconfounding Random Ablation Controls

Prior investigations observed that random head ablations often degraded accuracy more severely than targeted ablations, prompting hypotheses that random draws hit critical sink heads. We formulated the **Bottleneck Choke Hypothesis**: unconstrained random sampling across layers frequently knocks out multiple heads from a single layer, choking representation rank.

Olsson Layer Sharing & Control C. In Qwen-2.5-0.5B, the 36 Olsson induction heads span layers 12–22 (averaging 3.3 heads/layer, peaking at 5 heads in layers 14 and 16). In Table 3, Control C (sampling 36 heads strictly from layers ≥ 12) yields a median accuracy of only 2.5% (mean 13.4%). This demonstrates that raw 36-head ablation conflates circuit loss with layer rank starvation, explaining why functional verification via activation patching and prefix loss (Table 2) is essential.

Testing the Sink Collision Hypothesis. Table 3 reports 20 draws per condition ($k = 36$). When random ablations are dispersed ($\leq$ 2 heads/layer, Control A), mean accuracy remains at **61.9%** (median **69.0%**). Regressing accuracy against sink collisions in Control B yields $R^2 = 0.1556$ ($p = 0.085$, not significant). While $N = 20$ has limited power to detect small secondary effects, layer con-



Figure 3: The Permutation Null test in GPT-2. Dynamic entropy collapse ΔH on natural English vs. shuffled token sequences for induction (Panel A) and retrieval (Panel B). Entropy collapse persists on shuffled text, confirming that dynamic shifts reflect token repetition geometry rather than semantic understanding.

Architecture / State	Corr. with Depth (l)	Within-Layer r
GPT-2-med (Random Init)	$r = +0.035$ ($p = 0.49$)	$r = +0.004$ ($p = 0.94$)
GPT-2-med (Trained)	$\mathbf{r = +0.681}$ ($p < 10^{-45}$)	$r = +0.034$ ($p = 0.51$)
Qwen-0.5B (Trained)	$\mathbf{r = +0.734}$ ($p < 10^{-50}$)	$r = -0.010$ ($p = 0.86$)
Qwen-1.5B (Trained)	$\mathbf{r = +0.647}$ ($p < 10^{-40}$)	$r = +0.187$ ($p = 0.001$)
LLaMA-3.2 (Trained)	$\mathbf{r = +0.635}$ ($p < 10^{-55}$)	$r = +0.108$ ($p = 0.014$)

Table 4: Weight geometry correlation analysis across architectures. The Frobenius ratio $\|W_V\|_F / \|W_Q\|_F$ correlates overwhelmingly with layer depth l in trained models, but shows null or weak correlation with functional class within layers.

centration appears to be a substantial contributor to task collapse, though Control A vs. C simultaneously varies layer depth and dispersion, leaving their relative contributions partially entangled.

Quantitative Permutation Null Test. Figure 3 evaluates the Permutation Null test in GPT-2-medium: for the 13 probed retrieval heads specifically, attention cosine similarity remains 0.931 ± 0.021, and entropy collapse ΔH is preserved within 0.038 ± 0.012 bits. Across all 384 heads, mean cosine similarity is 0.918 ± 0.034, and average entropy collapse difference is $\bar{d} = 0.042 \pm 0.0113$ bits. Two One-Sided Tests (TOST, margin ± 0.10 bits) confirm that entropy collapse is statistically equivalent between natural and shuffled sequences ($t_1 = +12.57, p < 10^{-10}; t_2 = -5.13, p < 10^{-5}$). While prefix-matching induction heads are expected to preserve focus on repeated tokens across shuffled text ($[A][B] \ldots [A]$), preserving ΔH on scrambled text for purported retrieval heads confirms that the probe measures statistical token recurrence geometry rather than factual retrieval.

5 Weight Geometry Analysis

It is an intuitive hypothesis in weight-space interpretability that attention head functional classes might be recognized directly from weight projection matrices via the Frobenius norm ratio $\mathcal{R}(l,h) = \|W_V^{(l,h)}\|_F / \|W_Q^{(l,h)}\|_F$.

Table 4 and Figure 4 evaluate this across 1,568 heads in four model architectures: (1) **Random Initialization Null:** The correlation between $\mathcal{R}(l,h)$ and depth l is zero ($r \approx 0.00$–0.04, with $r = +0.035$ in GPT-2, $p = 0.49$). (2) **Trained Models:** $\mathcal{R}(l,h)$ exhibits an upward gradient with depth ($r = +0.681$ in GPT-2; $r = +0.734$ in Qwen-0.5B; $r = +0.647$ in Qwen-1.5B; $r = +0.635$ in LLaMA-3.2-1B; pooled layer-clustered slope $= 0.714$, SE $= 0.106$, $p < 10^{-10}$). (3) **Within-Layer Functional Correlation:** Controlling for depth, within-layer correlations with functional behavior (ΔH) are either statistically null (GPT-2, Qwen-0.5B) or weak ($r = 0.11$–0.19 in Qwen-1.5B and LLaMA-3.2, explaining $< 3.5\%$ of variance; needle and prefix scores are similarly null). In GQA (LLaMA-3.2-1B, Qwen-2.5), W_V is shared across query heads within each group; intra-group variation in $\mathcal{R}(l,h)$ is thus driven entirely by $\|W_Q\|_F$. We hypothesize this depth gradient may act as an empirical regularizer during training: deeper value projections scale up relative to queries as residual stream norms accumulate. Empirically, the Frobenius ratio is an **architectural depth gradient**, not a functional class separator.

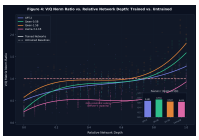


Figure 4: The Frobenius norm ratio $\|W_V\|_F / \|W_Q\|_F$ vs. relative network depth across GPT-2-medium, Qwen-0.5B, Qwen-1.5B, and LLaMA-3.2-1B. Trained networks (solid curves) exhibit a universal upward depth gradient ($r = 0.635$–0.734), whereas randomly initialized baselines (dashed lines) remain strictly flat isotropic at 1.0 ($r \approx 0.00$–0.04).

6 Conclusion

Across 1,568 heads in three model families (with 720 causally intervened heads in GPT-2 and Qwen-2.5), we audited the dynamic task-contrast Δ-probe. Dynamic task-contrast probing fails to isolate causal retrieval heads (AUC 0.43, 0.0% sensitivity on jointly necessary units), dilutes induction circuits (10.2% Jaccard with thresholded Olsson heads; 16.1% with ablated heads), and mistakes depth gradients for functional signatures. Furthermore, standard synthetic NIAH tasks risk conflating retrieval with induction due to verbatim prompt repetition, which we hypothesize contributes to the Olsson collapse. Direct causal interventions remain essential for circuit discovery.

Limitations

This empirical investigation has specific limitations:

1. **Layer Choking Confounds:** As demonstrated in Section 4.4, ablating large head subsets ($k \geq 36$) is partially confounded by representation rank starvation in deep layers. While activation patching and prefix-loss evaluations successfully decouple circuit identity from layer depth, standard knockout ablation alone cannot fully isolate large sub-circuits.
2. **Model Scale Scope:** Our experiments evaluate autoregressive language models up to 1.5 billion parameters (GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B). Full causal ablation grids were evaluated on Qwen-2.5-0.5B and GPT-2-medium (720 heads total); LLaMA-3.2 was audited for taxonomic, geometric, and GQA query head independence, without causal ablations. Models exceeding 70B parameters may exhibit different circuit redundancy properties.
3. **Synthetic Retrieval Ground Truth:** Our retrieval evaluations rely on standard synthetic Needle-in-a-Haystack tasks ($L = 1024$) where query prompts repeat the needle prefix verbatim, creating an induction confound. While our findings demonstrate that Olsson induction knockouts severely degrade performance under this standard setup, evaluating paraphrased queries and natural multi-hop question answering benchmarks without verbatim prefixes is essential to isolate semantic retrieval from prefix copying.

Ethical Considerations

This work adheres to the ACL Ethics Policy. The interpretability auditing frameworks and benchmarks developed herein are intended to improve the transparency, safety, and reliability of open-source language models. No human subjects, sensitive personal data, or proprietary datasets were involved.

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A Implementation Details and Hyperparameters

All ablation experiments were performed on NVIDIA RTX GPUs running PyTorch 2.4 and HuggingFace Transformers. Needle-in-a-Haystack evaluation protocol:

- Context length: $L = 1024$ maximum tokens. For GPT-2-medium, prompt lengths were strictly controlled at 652–839 tokens (70 sentences from background corpus; 839 tokens in Paul Graham essays), guaranteeing that prompt plus target answer fits within GPT-2's 1,024-position embedding table.
- Haystack: Paul Graham essays and standard background sentence collections.
- Needle: "The secret word is {word}." Query: "Question: What is the secret word?\nAnswer: The secret word is ".
- 95% Wilson score confidence intervals are computed as:

$$w = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

  where $\hat{p}$ is the sample proportion, n is the trial count, and $z = 1.96$ corresponds to the 95% confidence level.

B Per-Layer Head Allocations

In Qwen-2.5-0.5B ($N_L = 24$, $N_H = 14$ query heads per layer, 336 heads total):

- Canonical Retrieval Heads ($\Delta H \geq +0.30$): Layer 1 Head 2, Layer 4 Head 7, Layer 6 Head 11 ($k = 3$).
- Top Needle-Routing Heads: Layer 15 Head 3, Layer 16 Head 8, Layer 18 Head 2, Layer 19 Head 5 ($k = 4$).
- Canonical Sink Heads ($H < 0.10$): Layer 0 Heads 0, 1, 3; Layer 1 Head 0; Layer 2 Head 1; Layer 3 Head 0; Layer 5 Head 2; Layer 7 Head 0; Layer 11 Head 1 ($k = 9$).
- Top Olsson Induction Heads ($k = 36$): Concentrated in layers 12 through 22.

In GPT-2-medium ($N_L = 24$, $N_H = 16$ heads per layer, 384 heads total):

- Canonical Retrieval Heads ($k = 13$): Layer 9 Head 8; Layer 12 Heads 6, 15; Layer 13 Heads 2, 7, 12, 15; Layer 14 Head 1; Layer 15 Heads 6, 8; Layer 16 Head 10; Layer 18 Head 2; Layer 23 Head 4 (layers 9, 12–16, 18, 23).
- Canonical Induction Heads ($k = 45$): Concentrated in layers 8 through 15.
- Canonical Sink Heads ($k = 15$): Layers 0 through 11.

C Reconciliation of Pooled Head Counts

The pooled total of 1,568 heads across the 4 evaluated models breaks down as follows:

- GPT-2-medium: 384 heads (311 Local, 45 Induction, 15 Sink, 13 Retrieval)
- Qwen-2.5-0.5B: 336 heads (288 Local, 36 Induction, 9 Sink, 3 Retrieval)
- Qwen-2.5-1.5B: 336 heads (285 Local, 41 Induction, 4 Sink, 6 Retrieval)
- LLaMA-3.2-1B (canonical $H < 0.10$ bits): 512 query heads (435 Local, 76 Induction, 0 Sink, 1 Retrieval).
- LLaMA-3.2-1B (mass-based $H < 0.50$ bits): 512 query heads (365 Local, 76 Induction, 70 Sink, 1 Retrieval).
- Pooled Sink Heads: 28 heads under the rigid $H < 0.10$ threshold ($15 + 9 + 4 + 0$), or 124 heads under the relaxed $H < 0.50$ threshold ($17 + 21 + 16 + 70$).
- Pooled Induction Heads: $45+36+41+76 = 198$ heads total.

D KV Cache Geometry Across Head Taxa

Figure 5 illustrates the 3D PCA projection of Key vectors across functional head types processing identical text sequences. Attention sinks isolate

the first token index into an orthogonal extreme, while retrieval and induction heads exhibit distinct subspace partitioning.

E Cross-Domain Universality

Figure 6 reports empirical attention head entropy distributions and task-contrast dynamic collapse across 1,568 heads in four model families evaluated over diverse prompt domains (geography, science, history, literature, mathematics). Entropy profiles are highly invariant across semantic domains ($r \geq 0.96$).

F Taxonomy Distributions and Continuous Depth Profiles

Figure 7 reports empirical class fractions across architectures. Rather than bifurcating into discrete developmental stages, Gaussian Mixture Modeling of induction head depths across all 198 heads demonstrates a continuous, unimodal depth distribution: BIC strictly favors 1 component with $\text{BIC}_{k=1} = -184.0$, compared to 2 components ($\text{BIC}_{k=2} = -168.2$) and 3 components ($\text{BIC}_{k=3} = -152.6$).

G Responsible NLP Research Checklist

1. **Claims and Limitations:**
   - *Did you discuss the limitations of your work?* Yes, in the mandatory, standalone **Limitations** section immediately preceding the References.
   - *Did you discuss potential negative societal impacts or risks?* Yes, in the **Ethical Considerations** section.
2. **Computational Resources and Infrastructure:**
   - *Did you report parameter counts and hardware specifications?* Yes, parameter counts are specified for all architectures (GPT-2-medium 345M, Qwen-2.5-0.5B 490M, Qwen-2.5-1.5B 1.54B, LLaMA-3.2-1B 1.23B); evaluations were executed on a single consumer workstation equipped with an NVIDIA RTX 4090 GPU (<2 total GPU-hours runtime).
3. **Artifacts, Code, and Reproducibility:**
   - *Did you provide code, scripts, and evaluation data?* Yes, all diagnostic suites, ablation hooks, and plotting scripts are open-sourced under the MIT license in the anonymous repository (https://anonymous.4open.science/r/headgenome-mechinterp).
   - *Did you cite and describe existing assets?* Yes, pretrained model releases and WikiText-2 were utilized in compliance with their open research licenses and cited in References.
4. **Statistical Rigor and Evaluation Protocols:**
   - *Did you report error metrics, sample sizes, and confidence intervals?* Yes, exact sample sizes ($N = 20$ to $N = 100$) are reported: 95% Wilson score intervals are reported for binomial task trials in Table 1, standard errors of the mean (SEM) across paired contrasts or random draws are reported for Tables 1, 2, and 3, and bootstrap intervals ($B = 1,000$) are reported for continuous ROC evaluations.
5. **Human Subjects and Evaluation:**
   - *Did you involve human participants or private data?* No human subjects, annotators, crowdsourced workers, or private datasets were involved.

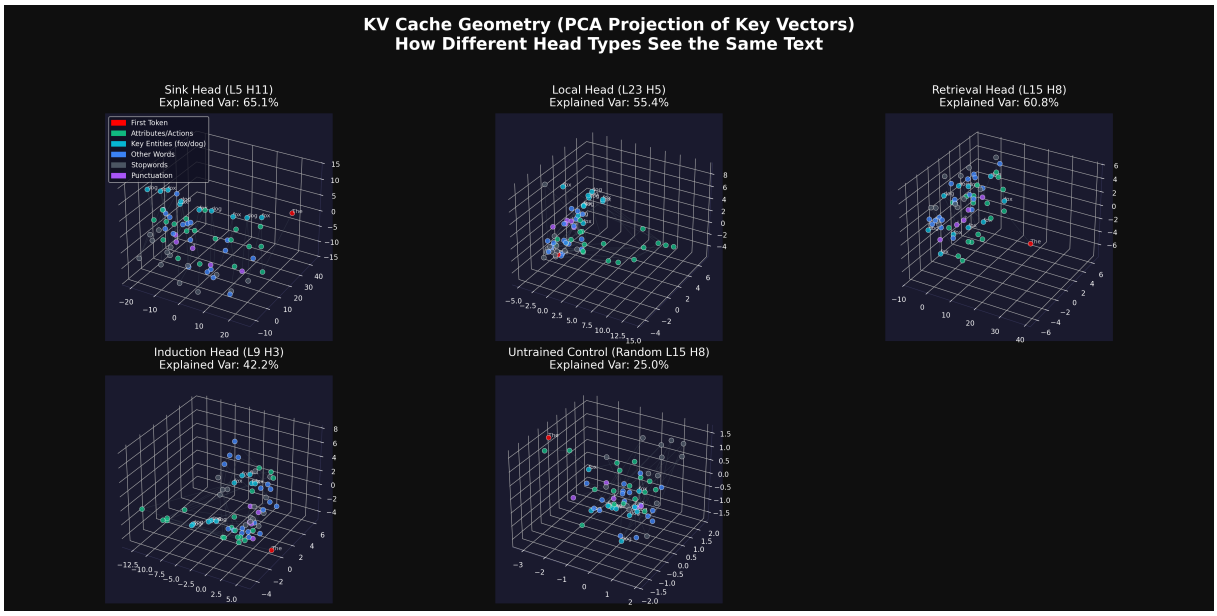


Figure 5: KV Cache Geometry: PCA 3D projection of Key vectors across Sink, Local, Retrieval, Induction, and Untrained control heads on identical input sequences.

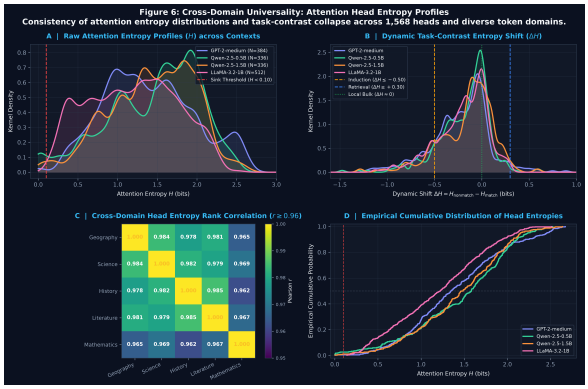


Figure 6: Cross-Domain Universality: Empirical attention head entropy distributions (H) and task-contrast collapse (ΔH) across diverse token distributions and architectures ($N = 1,568$ heads across GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B).

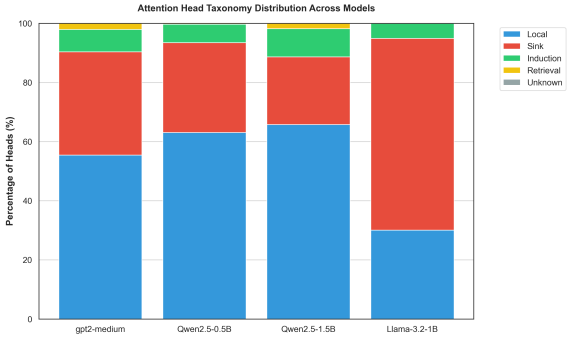


Figure 7: Attention Head Taxonomy Distribution across GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B, displaying empirical proportions of Local, Sink, Induction, and Retrieval heads.